

AI/ML-Based Predictive Maintenance and Vehicle Health Monitoring System

1. Introduction

Predictive maintenance uses vehicle sensor data and machine learning to identify abnormal operating conditions before they lead to component failure. This project develops an AI/ML-based predictive maintenance and vehicle health monitoring system for electric vehicles (EVs).

The system uses telemetry such as battery voltage, battery temperature, motor temperature, motor vibration, motor RPM, torque, power consumption, brake pressure, driving speed, distance travelled, and charge cycles. These parameters are processed and given to machine learning and deep learning models to estimate the probability of failure.

The implemented system compares **Logistic Regression, Random Forest, XGBoost, and LSTM** models. The final system converts the estimated failure probability into a vehicle health score and generates a maintenance status.

2. Objective

The main objectives of this project are:

- To analyse **EV telemetry data** for predictive maintenance.
 - To use important vehicle parameters for **failure prediction**.
 - To compare Logistic Regression, Random Forest, XGBoost, and LSTM.
 - To evaluate the models using accuracy, precision, recall, F1-score, and ROC-AUC.
 - To identify important vehicle parameters using XGBoost feature importance.
 - To convert model-estimated failure probability into a vehicle health score.
 - To generate **maintenance alerts** based on the **estimated vehicle condition**.
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3. Dataset Description

The project uses the [EVIoT Predictive Maintenance Dataset](#), obtained from Kaggle. The dataset contains **175,393 records and 30 columns** of EV telemetry data.

The data includes battery, motor, braking, tire, environmental, driving, and maintenance-related information.

The timestamped records allow the data to be analysed as a time series, which is particularly useful for the LSTM model.

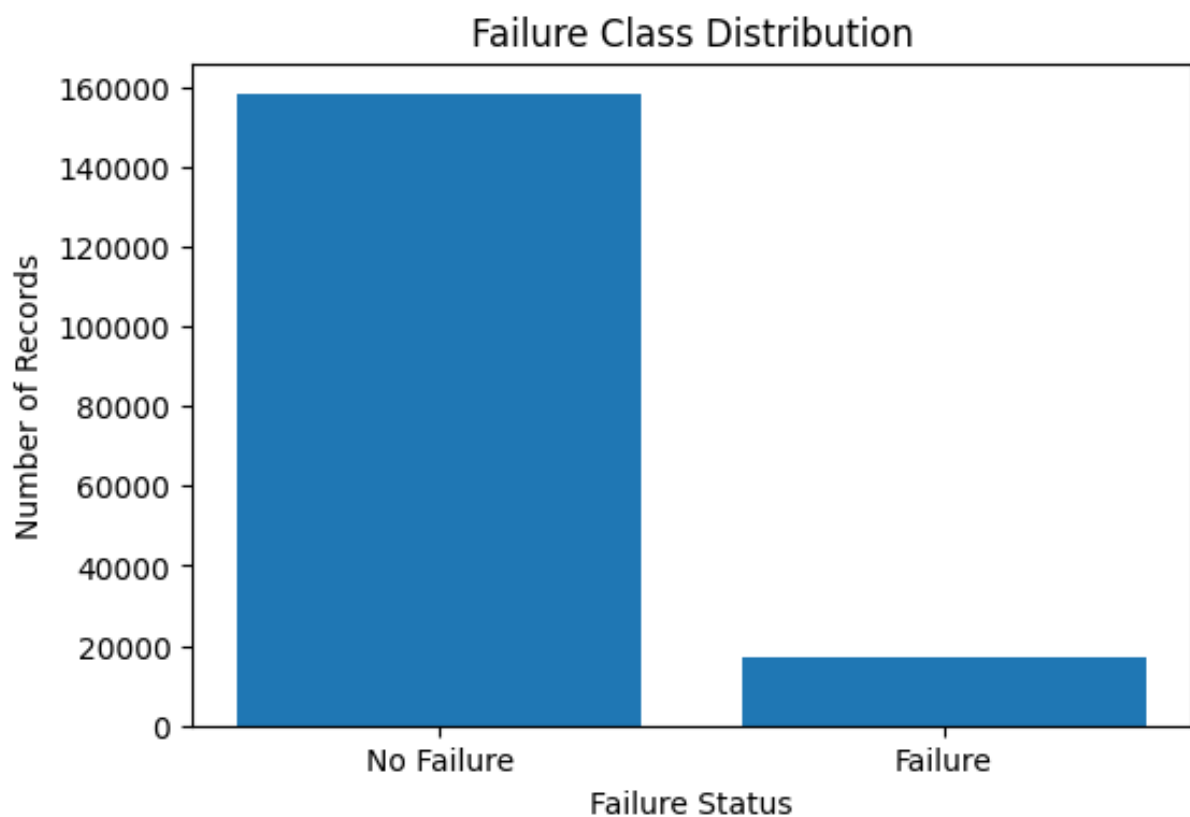
Table 1. shows the main Predictive Features selected for this project.

CATEGORY	SELECTED PARAMETERS
Battery	Battery Voltage, Battery Current, Battery Temperature
Motor	Motor Temperature, Motor Vibration, Motor RPM, Motor Torque
Energy	Power Consumption
Braking	Brake Pressure
Driving	Driving Speed, Distance Travelled
Usage	Charge Cycles

The target variable is **Failure_Probability**, where 0 represents no failure and 1 represents a failure condition.

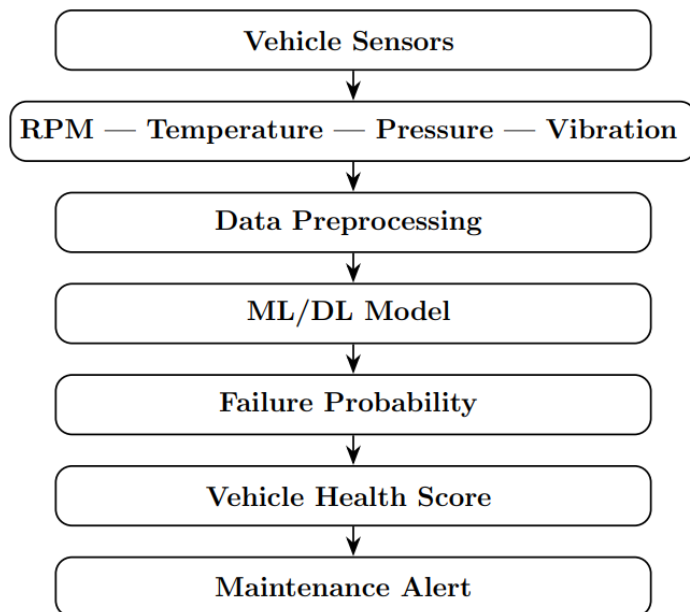
The dataset contained **no missing values**.

The failure distribution was approximately **90% non-failure and 10% failure**, providing substantially more failure examples than the initial dataset considered for this project.

Figure 1. Failure Class Distribution – how many normal vs failure records are present.

4. Methodology

System workflow:



5. Data Preprocessing

The Timestamp column was converted into datetime format and the dataset was sorted chronologically. The selected sensor parameters were separated from the target variable.

To prevent future observations from being used during training, the dataset was divided chronologically into **80% training data and 20% testing data**.

The resulting datasets contained:

- Training: **140,314 records**
- Testing: **35,079 records**

StandardScaler was fitted on the training data and then applied to the testing data for Logistic Regression and LSTM. Random Forest and XGBoost were trained using the original feature values.

Because failure records were less frequent than normal records, class weighting was used for Logistic Regression and Random Forest, while XGBoost used *scale_pos_weight*. For the LSTM, balanced class weights were calculated from the training sequences.

For temporal learning, the LSTM used **10 consecutive observations** as one input sequence. Since each observation represents a 15-minute interval, each sequence represents **2.5 hours of vehicle telemetry**.

6. Machine Learning Models

Four models were implemented.

6.1 Logistic Regression

Logistic Regression was used as a baseline classification model to establish a simple reference performance.

6.2 Random Forest

An ensemble of decision trees capable of learning non-linear relationships between vehicle parameters and failure conditions.

6.3 XGBoost

A gradient-boosting model used for classification. Class weighting was applied to give greater importance to the less frequent failure class.

6.4 LSTM

A recurrent neural network designed for sequential data. It uses consecutive vehicle telemetry observations to learn temporal patterns that may indicate changing vehicle conditions.

7. Model Results

Table 2. shows Accuracy, Precision, Recall, F1-score, and ROC-AUC evaluations for each model.

MODEL	ACCURACY	PRECISION	RECALL	F1-score	ROC-AUC
Logistic Regression	0.5475	0.0979	0.4334	0.1598	0.4932
Random Forest	0.9007	0.0000	0.0000	0.0000	0.5068
XGBoost	0.6163	0.1047	0.3797	0.1642	0.5162
LSTM	0.6501	0.1013	0.3205	0.1539	0.5037

The comparison shows how traditional machine learning models and a sequence-based deep learning model perform on the same predictive maintenance task. Accuracy is considered together with recall, F1-score, and ROC-AUC because the failure class is smaller than the non-failure class.

Figure 2. ROC Curve Comparison of ML and DL Models.

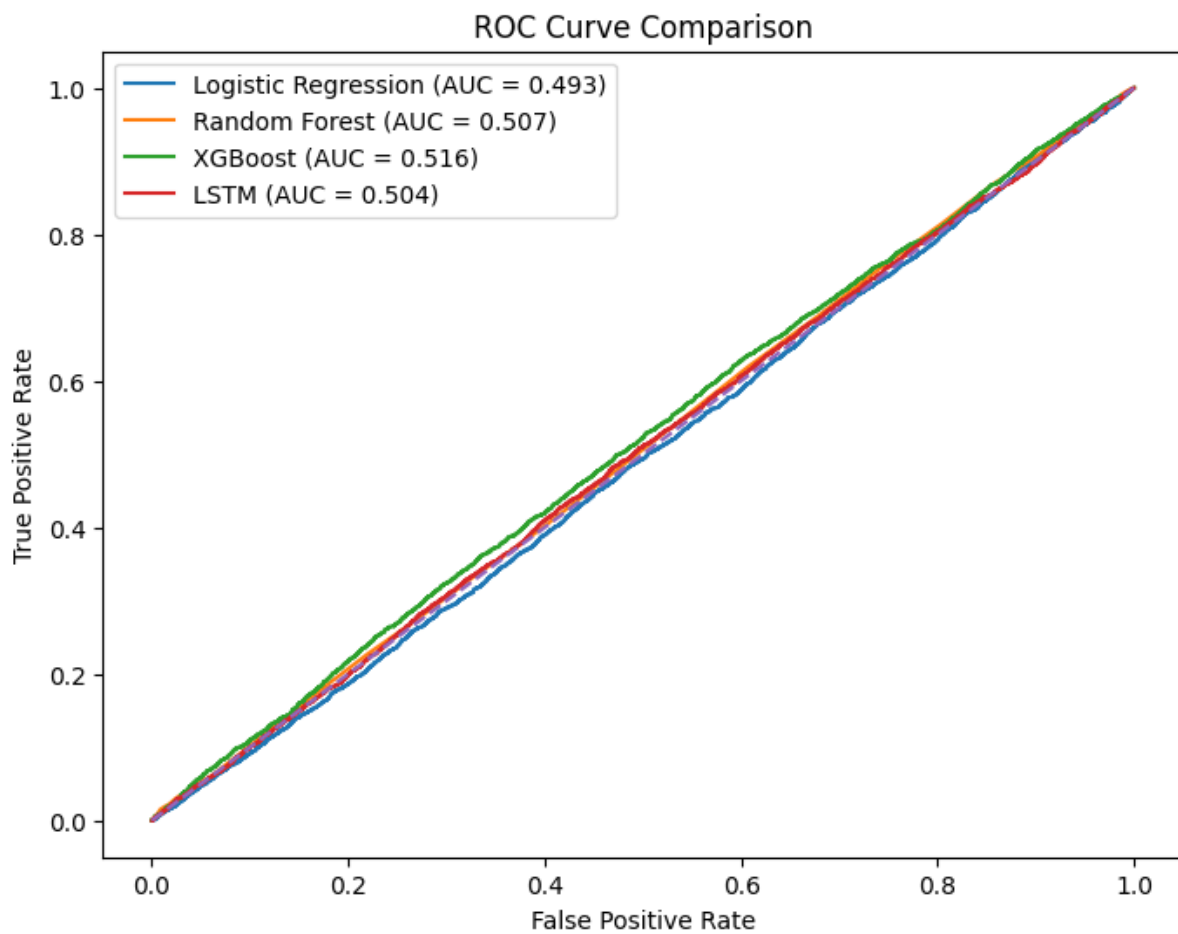
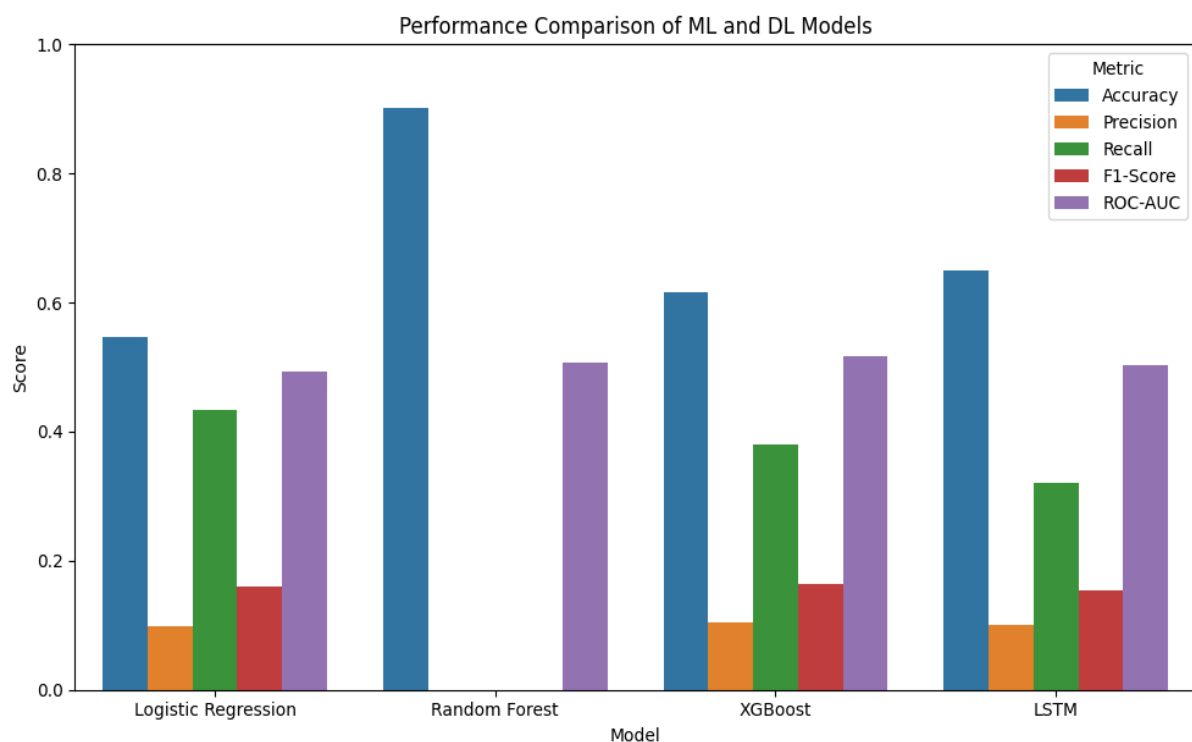


Figure 3. Performance Comparison of ML and DL Models.



The confusion matrices provide an additional view of correctly and incorrectly classified failure and non-failure records.

Figure 4: Confusion Matrices of the following models:

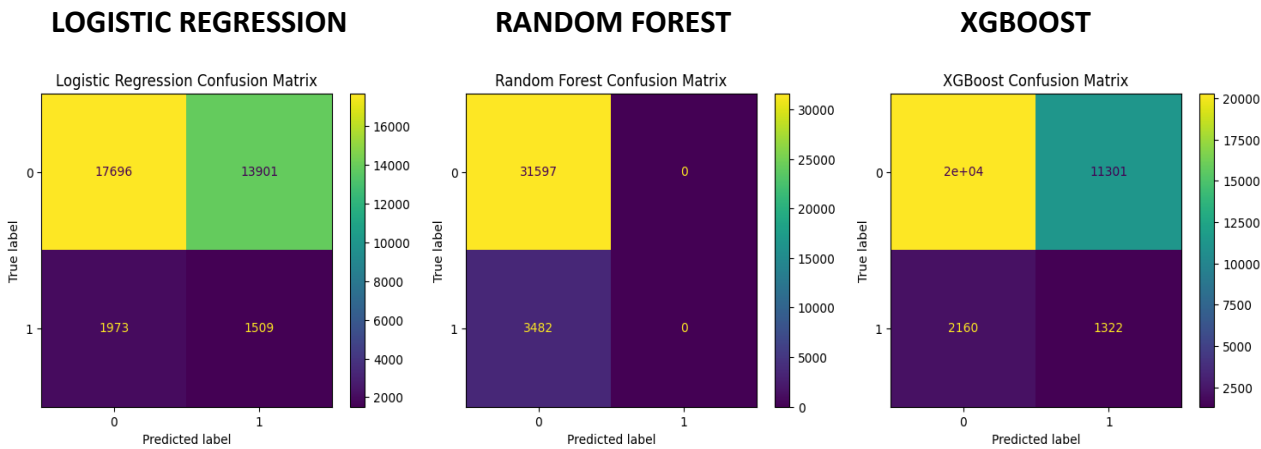
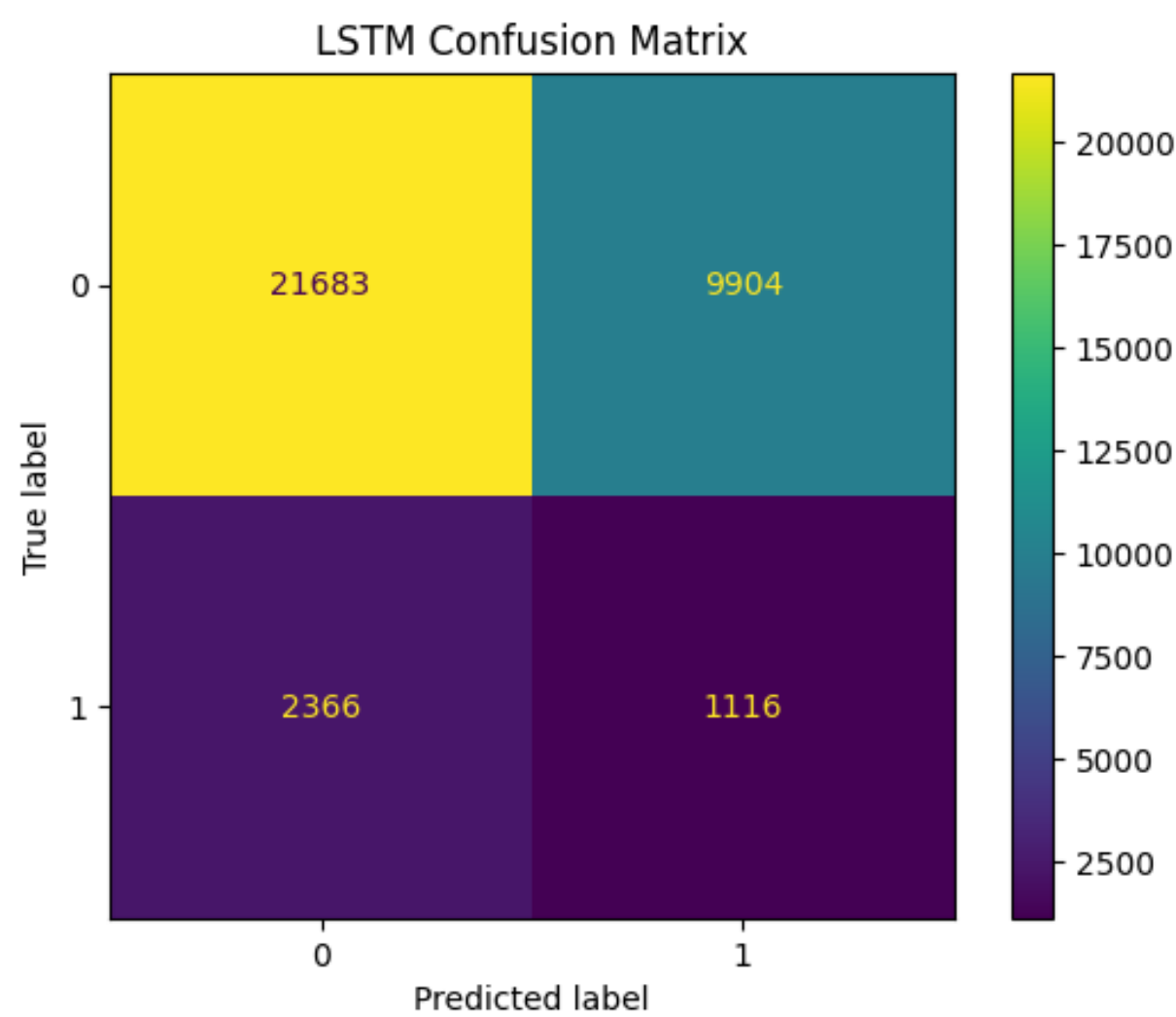


Figure 5: Confusion Matrix of LSTM model.



8. Vehicle Health Monitoring

The trained XGBoost model is used as the predictive component of the vehicle monitoring system. For each test observation, the model produces an estimated failure probability.

A health score is calculated as:

$$\text{Vehicle Health Score} = (1 - \text{Estimated Failure Probability}) \times 100$$

Table 3. shows all the Thresholds used in this project.

HEALTH SCORE	STATUS	MAINTENANCE ACTION
$\geq 70\%$	NORMAL	No immediate maintenance alert
40–69.99%	WARNING	Maintenance inspection recommended
$< 40\%$	CRITICAL	Immediate maintenance attention required

These thresholds are **project-defined thresholds** and are not intended to represent industry-standard limits.

Figure 6. Real-Time Vehicle Health Score, Failure Probability, and Maintenance Status.

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... Vehicle Health Monitoring System
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Vehicle Health: 49.37%
Failure Probability: 50.63%
Status: WARNING
Maintenance Alert: Maintenance inspection recommended

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The output displays the estimated failure probability, vehicle health score, status, and corresponding maintenance recommendation.

9. Explainability

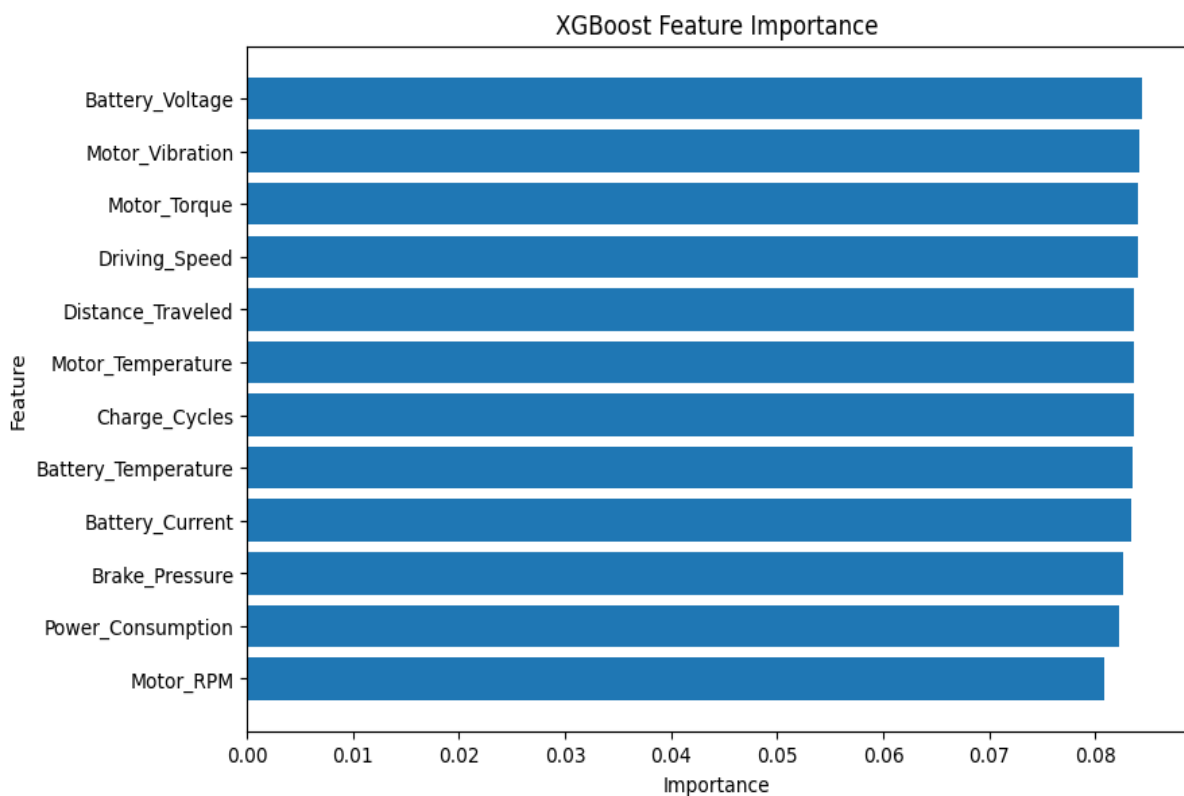
XGBoost feature importance was used to identify which vehicle parameters contributed most to the model's predictions.

Table 4. shows Importance values of XGBoost Feature.

INDEX	FEATURE	IMPORTANCE
0	Battery_Voltage	0.084442
4	Motor_Vibration	0.084166
6	Motor_Torque	0.084001
9	Driving_Speed	0.083977

10	Distance_Traveled	0.083622
3	Motor_Temperature	0.083582
11	Charge_Cycles	0.083579
2	Battery_Temperature	0.083544
1	Battery_Current	0.083433
8	Brake_Pressure	0.082651
7	Power_Consumption	0.082178
5	Motor_RPM	0.080825

Figure 7. XGBoost Feature Importance for Vehicle Failure Prediction.



The feature-importance results help interpret which telemetry variables the model relies on when making predictions. These values represent **model importance** and **should not be interpreted as proof of direct causation**.

9. Discussion

The project demonstrates a **complete predictive-maintenance pipeline**, starting from **EV telemetry** and ending with a **vehicle health and maintenance recommendation**.

The dataset provides a substantially larger number of observations than the initial dataset considered for the project and contains a broader range of EV-specific parameters. The **chronological train-test split** also provides a **more realistic evaluation setup** for time-dependent vehicle telemetry.

The comparison of **Logistic Regression, Random Forest, XGBoost, and LSTM** allows both conventional machine learning and deep learning approaches to be evaluated. The feature-importance analysis further provides an interpretable view of the parameters used by the XGBoost model.

However, model performance must be interpreted carefully because **the dataset is still imbalanced** and is a prepared predictive-maintenance dataset rather than live vehicle sensor data. The estimated failure probability represents the model's prediction on this dataset and should not be treated as a guaranteed real-world probability.

9. Limitations

- The dataset is based on EV telemetry rather than live data collected from the user's vehicle.
- The failure classes are imbalanced.
- The dataset's predefined failure-related variables may not represent every real-world vehicle failure scenario.
- Model-estimated probabilities are not necessarily calibrated real-world probabilities.
- XGBoost feature importance indicates the features used by the model but does not establish causation.
- The health-score thresholds used in the monitoring system are project-defined.

10. Conclusion

This project developed an **AI/ML-based predictive maintenance and vehicle health monitoring system** using EV telemetry data. The system processes multiple battery, motor, braking, driving, and usage parameters and compares Logistic Regression, Random Forest, XGBoost, and LSTM models for failure prediction.

The final monitoring pipeline converts the model's estimated failure probability into a vehicle health score and generates a maintenance status. XGBoost feature importance also provides an interpretable view of the vehicle parameters used by the predictive model.

Overall, the project demonstrates how machine learning and time-series deep learning can be integrated into a vehicle health monitoring pipeline to support **early identification of potential failure conditions and maintenance planning**.

11. References

- [1] EVIoT Predictive Maintenance Dataset, Kaggle.
- [2] Scikit-learn Developers, *Logistic Regression Documentation*.
- [3] Scikit-learn Developers, *Random Forest Classifier Documentation*.
- [4] XGBoost Developers, *XGBoost Documentation*.
- [5] TensorFlow Developers, *LSTM Layer Documentation*.